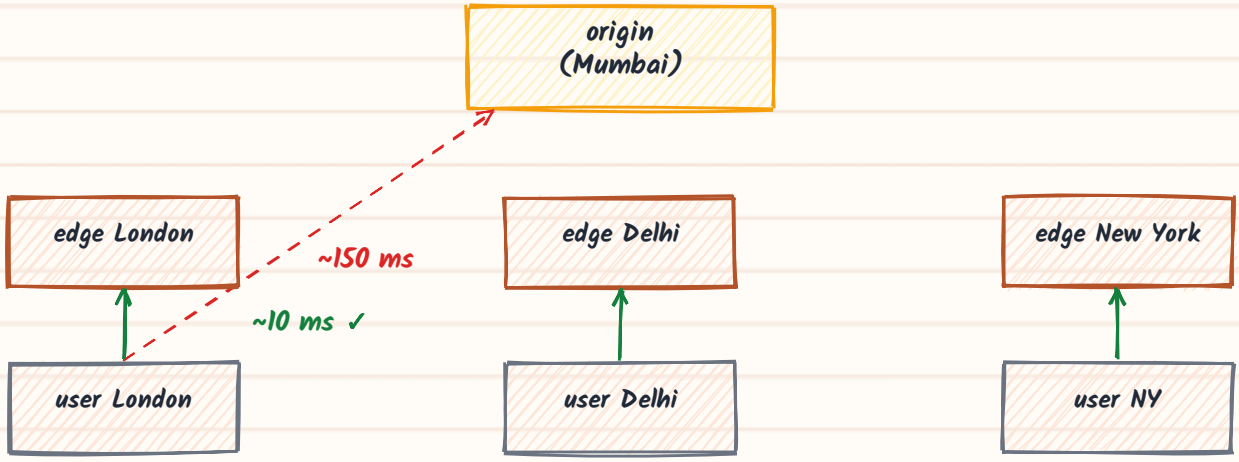
CDN & Edge Caching

Lesson 8 · how CDNs make every site fast

In this lesson → ① Slow photos ② What a CDN is ③ Pull CDN ④ Push vs pull ⑤ Cache headers ⑥ Invalidation ⑦ More than static ⑧ SnapLink + CDN ⑨ Interview Q&A
Running project: SnapLink, our photo-sharing app that grows from 1 server to 100M users.

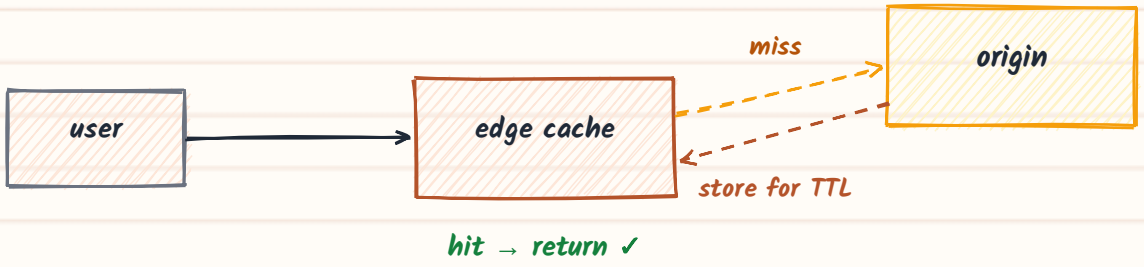
1 What a CDN is

> **CDN** = network of edge servers near users that cache content from your **origin**: lower latency, less origin load.



→ Examples: Cloudflare, Akamai, Amazon CloudFront, Fastly
→ Hundreds of cities (points of presence)

2 Pull CDN



3 Push vs pull

	Pull	Push
How	edge fetches from origin on first miss	you upload to the CDN in advance
Effort	low, self-managing	you manage what's stored
First request	slower (miss)	fast
Best for	many files, long tail	large, known-popular files

4 Cache headers

Header	Meaning
Cache-Control: public, max-age=N	any cache may keep it N seconds
Cache-Control: private / no-store	browser only / never cache
ETag + If-None-Match	unchanged → 304 Not Modified
s-maxage	TTL for shared caches (CDN) only

5 Invalidation

Versioned file names	logo.a1b2c3.png → new name on change *best*
Short TTL	Cache-Control: max-age=60 *fresh, more origin load*
Purge API	invalidate /img/* *urgent fixes*

Best practice
Put a content hash in the file name (app.3f9a.js) and cache for a year. A change = a new URL.

6 More than static files

- ✓ TLS termination near users
- ✓ DDoS protection and WAF
- ✓ Dynamic acceleration over the CDN backbone
- ✓ Edge functions: redirects, A/B tests, auth checks
- ✓ Video: caches inside ISPs (e.g. Netflix Open Connect)

7 SnapLink with a CDN

Content	Strategy
Photos, thumbnails	cache long, versioned URLs
JS / CSS / fonts	cache long, hashed names
Personal feed API	pass through (TLS + DDoS only)
Public trending page	cache briefly (e.g. 30 s)

Origin shield
A middle cache layer between edges and origin so a miss storm doesn't hit the origin from every edge.

→ Bandwidth savings = big cost savings

→ Measure the cache hit ratio (aim 90%+ for static)

8 Interview Corner

Q. What is a CDN?

A. A network of edge servers near users that cache content from the origin, reducing latency and origin load.

Q. Push vs pull CDN?

A. Pull fetches from origin on a miss; push requires uploading content in advance.

Q. How do you invalidate CDN content?

A. Prefer versioned (hashed) file names; otherwise short TTLs or a purge API.

Q. Can you cache API responses on a CDN?

A. Only public/shared responses with proper headers; personal data should pass through.

Q. What is an origin shield?

A. An intermediate cache tier that collapses misses from many edges into one origin request.

⚡ Revise in 60 seconds

- → CDN = **edges near users**
- → **Pull**: fetch on miss
- → **Push**: upload ahead
- → Cache-Control, ETag, 304
- → **Versioned names** > purge
- → Origin shield
- → TLS, DDoS, edge functions
- → Personal APIs pass through

Next lesson → SD-09
API design: REST, gRPC, GraphQL, WebSockets, SSE.
Watch on YouTube: **@TheCodeNotebook** · Notes: **thecodenotebook.com**